lived in the oceans. As the supercontinent Gondwana drifted slowly over the South Pole, a giant ice cap formed, lowering global temperatures. Much of the planet's water became "locked up" as ice, depressing sea levels and reducing the area of Earth's surface covered by ocean.

As a result, marine organisms living in shallow continental-shelf water suffered particularly high rates of extinction. In at least two peak die-off periods, separated by hundreds of thousands of years, nearly 85 percent of marine species died out, including brachiopods, bryozoans, trilobites, graptolites, and echinoderms.

Slow extinction

By the Late Devonian period, around 359 million years ago, the continents had been colonized by plants and insects, and massive organic reefs thrived in the oceans. The continents of Euramerica and Gondwana were converging into what would become Pangaea—the last of the supercontinents. In this period, there were two large extinction events, 374 and 359 million years ago. The former, occurring at the Frasnian-Famennian stage boundary, was the larger mass extinction.

The extinctions may have had many causes, including reduced oxygen in the oceans, falling sea levels, atmospheric changes, nutrient release by the spread of plants, and large-scale volcanic eruptions. Most organisms lived in the oceans, and shallow seas were worst affected, with many reef-building organisms, brachiopods, placoderms, and the last of the graptolite species dying off. Around 75 percent of marine species died, and it would be millions of years before corals reestablished themselves on a large scale.

"The current extinction has its own novel cause: not an asteroid or a massive volcanic eruption but "one weedy species.""

Elizabeth Kolbert

American journalist